

COLLEGE_DB-QUERY-DEMONSTRATION REPORT

1. PROJECT OVERVIEW:

This project demonstrates SQL querying techniques on a **Student Database**.

It covers:

- Basic retrieval of records
- Use of DISTINCT, ALIAS, WHERE, IN, BETWEEN, LIKE operators
- Arithmetic operations on columns
- Filtering with NULL checks and logical conditions

The goal is to showcase practical SQL skills for managing and analysing student and course data.

2. OBJECTIVES:

- Retrieve and display complete records from tables (Students, Courses, Enrolment).
 - Extract **unique values** using DISTINCT.
 - Improve readability with **column aliases**.
 - Apply **filtering conditions** (WHERE, IN, BETWEEN, LIKE).
 - Perform **arithmetic operations** on columns.
 - Handle **NULL values** effectively.
 - Demonstrate **comparison operators** (>, <, NOT IN).
-

3. TOOLS USED

- **Database:** MySQL
 - **Tables:**
 - Students (student details: name, age, city, status)
 - Courses (course name, credits)
 - Enrollment (student-course mapping with enrollment date)
-

4. KEY QUERIES & RESULTS:

Basic Retrieval

```
SELECT * FROM Students;
```

```
SELECT * FROM Courses;
```

```
SELECT * FROM Enrollment;
```

```

1  -- 1CREATE DATABASE
2
3  • CREATE DATABASE StudentsDatabase;
4  • USE StudentsDatabase;
5
6  -- CREATE TABLES
7
8  • CREATE TABLE Students (
9      StudentID INT PRIMARY KEY,
0      StudentName VARCHAR(50),
1      Age INT,
2      City VARCHAR(50)
3  );
4
5  • CREATE TABLE Courses (
6      CourseID INT PRIMARY KEY,
7      CourseName VARCHAR(50),
8      Duration VARCHAR(20)
9  );
0
1  • CREATE TABLE Enrollments (
2      EnrollmentID INT PRIMARY KEY,
3      StudentID INT,
4      CourseID INT,
5      EnrollmentDate DATE,
6      FOREIGN KEY (StudentID) REFERENCES Students(StudentID),
7      FOREIGN KEY (CourseID) REFERENCES Courses(CourseID)
8  );
9
0

```

DISTINCT VALUES:

SELECT DISTINCT city FROM Students;

```

5  • select
6      distinct city
7      from students;
8

```

city
Chennai
Bangalore
Delhi
Mumbai
Kolkata
Hyderabad
Pune
Unknown

SELECT DISTINCT city AS list_of_city FROM Students;

```
9 • select
0 distinct city as list_of_city
1 from students;
2 • select name as student name, age, city as location from
```

result Grid | Filter Rows: | Export: | Wrap Cell Content

list_of_city
Chennai
Bangalore
Delhi
Mumbai
Kolkata
Hyderabad
Pune
Unknown

ALIASES:

SELECT name AS student name, age, city AS location FROM Students;

```
12 • select name
13 as student_name,
14 age,
15 city as location from students;
16 • select * from students where city='chennai';
```

result Grid | Filter Rows: | Export: | Wrap Cell Content:

student_name	age	location
Arun Kumar	20	Chennai
Meena Raj	22	Bangalore
Rahul Sharma	21	Delhi
Priya Singh	23	Mumbai
Vijay Kumar	24	Kolkata
Sakshi Verma	19	Hyderabad
Rajat Singh	22	Pune
Neha Gupta	20	Chennai
Tanvi Sharma	23	Bangalore
Akash Mehta	21	Delhi
Riya Kapoor	22	Mumbai
Kavya Sharma	21	Unknown
Tanmay Singh	24	Unknown
Anjali Mehta	22	Unknown

FILTERING:

SELECT * FROM Students WHERE city = 'Chennai';

```
18 • select *
19   from students
20   where city='chennai';
21 • select * from students where age>22;
```

Result Grid | Filter Rows: | Edit: | Export/Import: |

StudentID	Email	Name	Age	Status	City
1	arun@gmail.com	Arun Kumar	20	Active	Chennai
8	neha@gmail.com	Neha Gupta	20	Active	Chennai
NULL	NULL	NULL	NULL	NULL	NULL

SELECT * FROM Students WHERE age > 22;

```
21 • select *
22   from students
23   where age>22;
24 • select name, age, age + 5 as age after 5yrs from students;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | W

StudentID	Email	Name	Age	Status	City
4	priya@gmail.com	Priya Singh	23	Active	Mumbai
5	vijay@gmail.com	Vijay Kumar	24	Active	Kolkata
9	tanvi@gmail.com	Tanvi Sharma	23	Inactive	Bangalore
13	tanmay@gmail.com	Tanmay Singh	24	NULL	Unknown
NULL	NULL	NULL	NULL	NULL	NULL

ARITHMETIC OPERATIONS:

SELECT name, age, age + 5 AS age_after_5yrs FROM Students;

```
24 • select
25   name, age,
26   age + 5 as age_after_5yrs
27   from students;
28 • select Credits*2 as DoubledCredits, CourseName from cou
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content

name	age	age_after_5yrs
Arun Kumar	20	25
Meena Raj	22	27
Rahul Sharma	21	26
Priya Singh	23	28
Vijay Kumar	24	29
Sakshi Verma	19	24
Rajat Singh	22	27
Neha Gupta	20	25
Tanvi Sharma	23	28
Akash Mehta	21	26
Riya Kapoor	22	27
Kavya Sharma	21	26
Tanmay Singh	24	29
Anjali Mehta	22	27

SELECT Credits * 2 AS Doubled Credits, Course Name FROM Courses;

```
8 • select
9 Credits*2 as DoubledCredits ,
0 CourseName
1 from courses;
2 • select
```

Result Grid | Filter Rows: | Export: | Wrap Cell C

DoubledCredits	CourseName
8	Data Science with Python
4	Advanced Excel
4	Excel Basics
10	Machine Learning
6	Power BI
10	Python for Data Analysis
8	SQL Fundamentals
6	Statistics for Data Analysis

COMPARISON OPERATORS;

SELECT course Name, credits FROM Courses WHERE Credits > 2;

```
32 • select
33 coursename,credits
34 from courses
35 where Credits >2;
36 • select
37 coursename,credits
38 from courses
```

Result Grid | Filter Rows: | Export: |

coursename	credits
Data Science with Python	4
Machine Learning	5
Power BI	3
Python for Data Analysis	5
SQL Fundamentals	4
Statistics for Data Analysis	3

SELECT course Name, credits FROM Courses WHERE Credits <> 2;

```
6 • select
7   courseName,credits
8   from courses
9   where Credits <>2;
```

SELECT *

Result Grid | Filter Rows: | Export: | Wrap Cell

courseName	credits
Data Science with Python	4
Machine Learning	5
Power BI	3
Python for Data Analysis	5
SQL Fundamentals	4
Statistics for Data Analysis	3

LOGICAL CONDITIONS:

SELECT * FROM Students WHERE age > 20 AND status = 'active';

```
40 • SELECT *
41   FROM Students
42   where age>20 and status = 'active';
43 • SELECT *
44   FROM Students
45   where status is null;
```

Result Grid | Filter Rows: | Edit: | Export/Import: |

StudentID	Email	Name	Age	Status	City
2	meena@gmail.com	Meena Raj	22	Active	Bangalore
3	rahul@gmail.com	Rahul Sharma	21	Active	Delhi
4	priya@gmail.com	Priya Singh	23	Active	Mumbai
5	vijay@gmail.com	Vijay Kumar	24	Active	Kolkata
10	akash@gmail.com	Akash Mehta	21	Active	Delhi
11	riya@gmail.com	Riya Kapoor	22	Active	Mumbai
12	NULL	Kavya Sharma	21	Active	Unknown
NULL	NULL	NULL	NULL	NULL	NULL

NULL HANDLING:

SELECT * FROM Students WHERE status IS NULL;

```
3 • SELECT *
4 FROM Students
5 where status is null;
6 • SELECT *
7 FROM Students
8 where status is not null;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

StudentID	Email	Name	Age	Status	City
13	tanmay@gmail.com	Tanmay Singh	24	NULL	Unknown
14	NULL	Anjali Mehta	22	NULL	Unknown
NULL	NULL	NULL	NULL	NULL	NULL

SELECT * FROM Students WHERE status IS NOT NULL;

```
47 • SELECT *
48 FROM Students
49 where status is not null;
50
51
```

Result Grid | Filter Rows: | Edit: | Export/Import:

StudentID	Email	Name	Age	Status	City
1	arun@gmail.com	Arun Kumar	20	Active	Chennai
2	meena@gmail.com	Meena Raj	22	Active	Bangalore
3	rahul@gmail.com	Rahul Sharma	21	Active	Delhi
4	priya@gmail.com	Priya Singh	23	Active	Mumbai
5	vijay@gmail.com	Vijay Kumar	24	Active	Kolkata
6	sakshi@gmail.com	Sakshi Verma	19	Active	Hyderabad
7	rajat@gmail.com	Rajat Singh	22	Inactive	Pune
8	neha@gmail.com	Neha Gupta	20	Active	Chennai
9	tanvi@gmail.com	Tanvi Sharma	23	Inactive	Bangalore
10	akash@gmail.com	Akash Mehta	21	Active	Delhi
11	riya@gmail.com	Riya Kapoor	22	Active	Mumbai
12	NULL	Kavya Sharma	21	Active	Unknown
NULL	NULL	NULL	NULL	NULL	NULL

IN / NOT IN

SELECT * FROM Students WHERE city NOT IN ('Delhi');

```
54 • SELECT *
55 FROM Students
56 where city not in('delhi');
57
58 • SELECT *
```

result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell C

StudentID	Email	Name	Age	Status	City
1	arun@gmail.com	Arun Kumar	20	Active	Chennai
2	meena@gmail.com	Meena Raj	22	Active	Bangalore
4	priya@gmail.com	Priya Singh	23	Active	Mumbai
5	vijay@gmail.com	Vijay Kumar	24	Active	Kolkata
6	sakshi@gmail.com	Sakshi Verma	19	Active	Hyderabad
7	rajat@gmail.com	Rajat Singh	22	Inactive	Pune
8	neha@gmail.com	Neha Gupta	20	Active	Chennai
9	tanvi@gmail.com	Tanvi Sharma	23	Inactive	Bangalore
11	riya@gmail.com	Riya Kapoor	22	Active	Mumbai
12	NULL	Kavya Sharma	21	Active	Unknown
13	tanmay@gmail.com	Tanmay Singh	24	NULL	Unknown
14	NULL	Anjali Mehta	22	NULL	Unknown
NULL	NULL	NULL	NULL	NULL	NULL

SELECT * FROM Students WHERE city IN ('Delhi', 'Chennai');

```
58 • SELECT *
59 FROM Students
60 where city in('delhi','chennai');
61 • SELECT *
```

StudentID	Email	Name	Age	Status	City
1	arun@gmail.com	Arun Kumar	20	Active	Chennai
3	rahul@gmail.com	Rahul Sharma	21	Active	Delhi
8	neha@gmail.com	Neha Gupta	20	Active	Chennai
10	akash@gmail.com	Akash Mehta	21	Active	Delhi
NULL	NULL	NULL	NULL	NULL	NULL

BETWEEN:

SELECT * FROM Students WHERE age BETWEEN 20 AND 24;

```
1
2 • SELECT *
3 FROM Students
4 where age between 20 and 24;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

StudentID	Email	Name	Age	Status	City
1	arun@gmail.com	Arun Kumar	20	Active	Chennai
2	meena@gmail.com	Meena Raj	22	Active	Bangalore
3	rahul@gmail.com	Rahul Sharma	21	Active	Delhi
4	priya@gmail.com	Priya Singh	23	Active	Mumbai
5	vijay@gmail.com	Vijay Kumar	24	Active	Kolkata
7	rajat@gmail.com	Rajat Singh	22	Inactive	Pune
8	neha@gmail.com	Neha Gupta	20	Active	Chennai
9	tanvi@gmail.com	Tanvi Sharma	23	Inactive	Bangalore
10	akash@gmail.com	Akash Mehta	21	Active	Delhi
11	riya@gmail.com	Riya Kapoor	22	Active	Mumbai
12	NULL	Kavya Sharma	21	Active	Unknown
13	tanmay@gmail.com	Tanmay Singh	24	NULL	Unknown
14	NULL	Anjali Mehta	22	NULL	Unknown
NULL	NULL	NULL	NULL	NULL	NULL

SELECT * FROM Enrollment WHERE enrollmentdate BETWEEN '2024-01-15' AND '2024-02-29';

```
66 • SELECT *
67 FROM enrollment
68 WHERE enrollmentdate BETWEEN '2024_01_15' AND '2024-02-29';
69
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

StudentID	CourseID	EnrollmentDate	Grade
1	PY201	2024-01-15	B
1	STAT301	2024-02-08	B
2	EXL201	2024-02-09	A
2	ML401	2024-02-01	A
3	PBI301	2024-01-18	C
3	STAT301	2024-02-02	B
4	DS501	2024-02-03	A
4	EXL102	2024-01-20	B
5	ML401	2024-01-25	B
5	SQL101	2024-02-05	C
6	DS501	2024-01-28	A
6	PY201	2024-02-06	B
7	EXL201	2024-01-30	C
7	PBI301	2024-02-07	A
NULL	NULL	NULL	NULL

LIKE OPERATOR:

SELECT * FROM Students WHERE name LIKE 'a%';

```
70 • SELECT *
71     FROM Students
72     where name like 'a%';
73
74 • SELECT *
```

Result Grid | Filter Rows: | Edit: | Export/Import:

StudentID	Email	Name	Age	Status	City
1	arun@gmail.com	Arun Kumar	20	Active	Chennai
10	akash@gmail.com	Akash Mehta	21	Active	Delhi
14	NULL	Anjali Mehta	22	NULL	Unknown
NULL	NULL	NULL	NULL	NULL	NULL

SELECT * FROM Courses WHERE course Name LIKE '%ANALYSIS%';

```
74 • SELECT *
75     FROM COURSES
76     WHERE COURSENAME LIKE '%ANALYSIS%';
77
```

Result Grid | Filter Rows: | Edit: | Export/Import:

CourseID	CourseName	Credits
PY201	Python for Data Analysis	5
STAT301	Statistics for Data Analysis	3
NULL	NULL	NULL

5. CONCLUSION:

This project demonstrates:

- **Core SQL operations** for data retrieval and filtering
- **Use of operators** (DISTINCT, ALIAS, WHERE, IN, BETWEEN, LIKE)
- **Arithmetic and comparison logic** in queries
- **Handling NULL values** for data quality checks

It highlights practical SQL skills for managing student and course information in a relational database.
